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ORIGINAL RESEARCH

A Delayed Evening Meal Enhances Sleep Quality in Young Rugby Players

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The aim of this study was to examine the effect of delayed evening mealtime on sleep quality in young athletes. Twelve rugby players (age 15.8 ± 0.7 years) participated in a crossover within-participant design. Adolescents spent five consecutive days in each of two conditions, separated by a 2-week washout period: routine dinner (3.5 hr before bedtime) and late dinner (LD, 1.5 hr before bedtime). Other mealtimes as well as bedtime and wake-up time were usual and remained the same in both conditions. Their schedules, dietary intakes, and physical activity were controlled and kept constant throughout the study. Sleep was assessed using polysomnography on the first and the last nights in the individual rooms of the boarding school. An increase in total sleep time by 24 min ($p = .001$, $d = 1.24$) and sleep efficiency by 4.8% was obtained during LD ($p = .001$, $d = 1.24$). Improvement in sleep efficiency was mainly due to a lower wake after sleep onset (–25 min, $p = .014$, $d = -3.20$), a decrease of microarousals (–25%, $p = .049$, $d = -0.64$), and awakenings ≥90 s (–30%, $p < .01$, $d = -0.97$) in LD compared to routine dinner. There were no significant differences in sleep architecture except for a shorter slow-wave sleep (N3) latency (–6.9 min, $p = .03$, $d = -0.778$) obtained during LD. In this study, evening dinner 1.5 hr before bedtime leads to better quality and less fragmented sleep compared to evening dinner 3.5 hr before bedtime in young athletes.

Keywords: dinnertime, sleep efficiency, polysomnography, adolescent athlete, chrononutrition

Irregular sleep–wake cycle, insufficient sleep duration, and poor sleep quality have been reported in athletes and particularly in youth (Fox et al., 2020). Young athletes get less total sleep duration and quality than senior and nonathletes (Saidi et al., 2022; Vlahoyiannis et al., 2020). Sleep has a restorative effect on immune and endocrine systems, is necessary to recover from the waking phase of the day, and plays a role in learning and memory (Doherty et al., 2019). Thus, impaired sleep induces detrimental effects on both physical and cognitive performances (Fullagar et al., 2015) and is likely to put young athletes at increased risk for injury and burnout (von Rosen et al., 2017).

While the primary role of nutrition is to meet the needs of tissue function, growth, adaptive effects of intense training, and recovery, it may also improve sleep (Gratwicke et al., 2021). The intake of specific foods that are rich in melatonin, tryptophan-rich protein, and micronutrients has beneficial effects on sleep. A longer sleep duration was reported with the consumption of tryptophan-rich protein (Doherty et al., 2019; St-Onge et al., 2016). Moreover, a shorter sleep latency was observed when high glycemic index carbohydrates were consumed at the evening meal (Gratwicke et al., 2021). However, Moss et al. (2022) found no significant relationship between sleep quality and consumption of dairy, fruit, and vegetables in athletes.

Beyond the quality and quantity of food, studies conducted in the field of chrononutrition have explored effects of frequency, regularity, and timing of food intake on the sleep–wake cycle. Indeed, the internal biological clock can be modified by changing not only the nature of the food but also the timing of food intake (Doherty et al., 2019). However, studies that have looked at the timing of food intake have focused on populations of interest with specific problems, such as shift workers (Nogueira et al., 2021). Recommendations on dinner timing suggest that dinner should be consumed 2–4 hr before bedtime (Fujiwara et al., 2005; Lopez-Minguez et al., 2018). In healthy adults, any food intake 30–60 min before bedtime is correlated with negative effects on sleep quality (Crispim et al., 2011). Conversely, there is evidence that eating close to bedtime may have positive effects on sleep. A recent study using polysomnography conducted on young adults demonstrated that shifting the dinner from 1 to 5 hr before habitual bedtime decreased deep sleep during the first half of the night (Duan et al., 2021). An association between the sleep quality score obtained by questionnaire and the response on eating a heavy meal in the 3 hr before bedtime was observed, indicating that eating close to bedtime is associated with longer total sleep duration (Knufinke et al., 2018). Similarly, Falkenberg et al. (2021) found, using accelerometry sleep measurement coupled with food diaries, that each additional hour between the main evening meal and bedtime resulted in shorter sleep duration (–8 min). These two observational studies are the only ones to have examined the effect of dinner time on sleep in young adult athletes. Thus, the effect of the timing of food intake on sleep characteristics remains poorly documented, especially in young athletes (St-Onge et al., 2016).

Therefore, the aim of the present study was to examine the effect of delayed evening mealtime on sleep quality in young athletes. We hypothesized that a shorter time slot between evening mealtime and bedtime would promote sleep quality.

Methods

Participants

Participants were recruited at Association Sportive Montferrandaise (ASM) Clermont Auvergne Rugby, France. All participants were

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living in the same boarding school. Inclusion criteria were male, age 14–18 years, engaged in competitive rugby. Exclusion criteria were obstructive sleep apnea syndrome (Berlin questionnaire; Netzer et al., 1999), smoking, use of medication susceptible to interfere with sleep outcomes, chronic illness, or injury at the time of the study.

Among the 14 participants assessed for eligibility, 12 rugby players aged 15.8 ± 0.7 years were included. All participants and their parents were informed about the study and provided written consent. The present research received ethical approval from a National Ethics Review Board (IRB 00012476-2021-20-01-79) and all experimental procedures were conformed to the principles of the Helsinki Declaration (Supplementary Material S2 [available online]).

Study Design

Effects of evening dinnertime on sleep quality were examined in free-living conditions using a crossover, within-participant design (Figure 1). The study was performed from May to June 2021 during a period in which school schedule and training load were constant (10 hr/week of training, between 14:00 and 18:30, same intensity and frequency, and competitions were cancelled due to COVID crisis). Eligible participants completed an inclusion visit including assessment of anthropometric characteristics, chronotype, and subjective sleep quality. One week prior to the experimental conditions launch, the investigators equipped the participants with a portable polysomnography (PSG) for a habituation night, and bedtime was fixed at 22:30 to avoid any effect of an irregular bedtime schedule on circadian rhythm and sleep quality during the study. Participants were randomized into two conditions (https://www.randomizer.org/): (a) routine evening dinner at 19:00 and (b) late evening dinner at 21:00. Other meals were administered at fixed times of the day: breakfast at 07:30, lunch at 12:00, snacks at 17:30. The two conditions were administered over five consecutive weekdays (from Monday morning to Friday evening) separated by a 2-week washout period. Sleep measurements were performed on Monday (D0) and Thursday night (D3). During each sleep measurement, participants were instructed to go to their room at 21:30. The investigators equipped them with polysomnography devices, and the light was switched off at 22:30. The evening use of electronic media was forbidden during the study. Bedtime and wake-up time were set at 22:30 and 07:00, respectively. Sleep measurements were carried out in the usual living conditions and in their own individual boarding room. During the 5 days of each testing sessions, physical activity (PA) of athletes was controlled using accelerometry, and participants were required to fill in daily activity and sleep diaries. On Session 1, participants self-selected their food from a panel of dishes offered at the school canteen. All foods were weighed by a dietician, prior to and post consumption using an electronic food scale. On Session 2, same food trays and snacks as in Session 1 were prepared and distributed to the participants. Participants were supervised to ensure full consumption of all meals. Athletes were asked not to have food intake between dinner and bedtime and not to consume sources of caffeine (e.g., coffee, energy drinks) throughout the study.

Measurements

Objective Sleep

Brain activity was monitored using an ambulatory multichannel electroencephalograph (Sleep Profiler PSG2, Advanced Brain

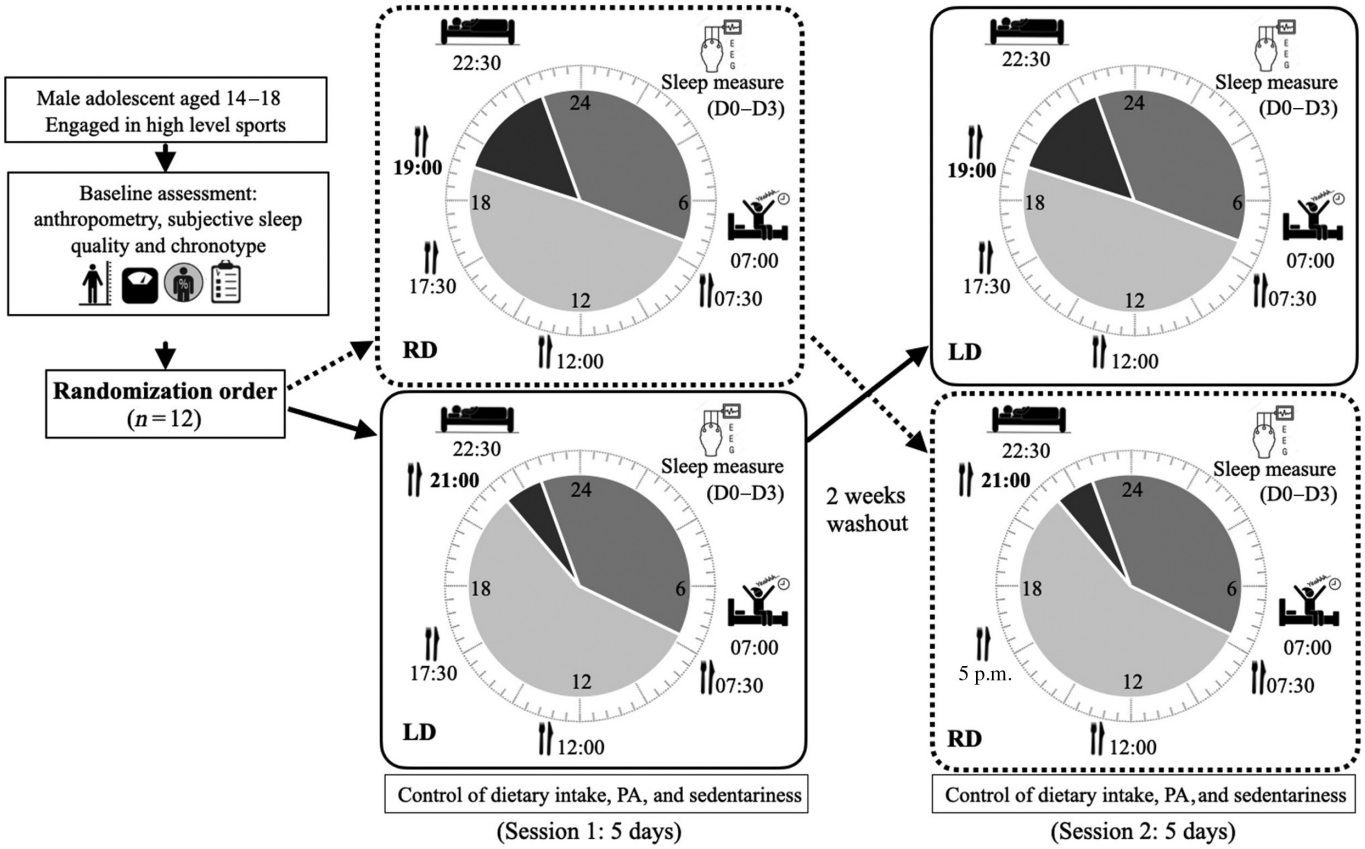

Figure 1 — Study design. *Note*. D0 = Monday (baseline); D3 = Thursday; PA = physical activity; RD= routine dinner; LD = late dinner.

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Monitoring). The system provides access to 13 channels: electroencephalography, electrooculography, and electromyography of front polar sites (two frontal electrodes on the forehead: AF7 and AF8; a reference sensor at the center of the forehead: Fpz) allowing the characterization of sleep staging and quality as well as wireless oximetry, nasal pressure/airflow, chest and abdomen respiratory effort, forehead and finger pulse rate, head movement and position, and quantitative snoring. Sleep records were downloaded from sleep profiler portal. Automated algorithms were applied to the signals from each 30-s epoch as recommended by the American Academy of Sleep Medicine (Berry et al., 2017). After processing data, an experienced sleep expert reviewed recordings from the polysomnography to confirm or amend the auto-sleep analysis if needed. A self-administered sleep diary was completed by participants during both sessions in order to confirm bedtime, wake-up time, and nighttime awakenings (Carney et al., 2012).

Self-Reported Sleep Quality

Subjective sleep quality and sleep disturbances were evaluated over the 4 weeks prior to the study from the Pittsburgh Sleep Quality Index (PSQI, Buysse et al., 1989). Scores from the PSQI are derived from the 19 items. The scores of the seven components were summed to yield a PSQI global score ranging from 0 to 21. Scores > 5 indicate "poor sleepers." The psychometric data of the French version show an acceptable internal homogeneity ($\alpha = .88$) and a test–retest reliability of 0.62 (Blais et al., 1997).

Chronotype

A French version of the Morningness–Eveningness Scale for Children (Carskadon et al., 1993) was used (Caci et al., 2005). This scale includes 10 items. Outcome scores range from 10 (eveningness) to 42 (morningness). Percentiles 10 and 90 are usually used as the cutoff points to differentiate the prototypical morning, intermediate, and evening types, equivalent to Morningness–Eveningness Scale for Children values of 10–19, 20–32, and 33–42, respectively. The internal validity of the 10 items is satisfactory: $\alpha = .76$ (Caci et al., 2005).

Dietary Intake

A nutritional analysis of dietary intakes weighed for 5 days on Sessions 1 and 2 was performed using an Excel spreadsheet based on the French food composition database (ANSES, 2020). Specific information on local foods (ingredients and amounts) has been added to the database. Energy intake (kcal) and macronutrients (g) were calculated for each meal of both sessions.

Physical Activity

Sedentary time and physical activity were measured using GT3X triaxial accelerometers (ActiGraph), completed by self-reported diaries. Participants were instructed to wear the accelerometer on their dominant wrist, except for water activities. Accelerometers were initialized for data collection at 30 Hz. Once raw data were downloaded, the ActiLife software version 6.13.4 (ActiGraph) was used to convert raw data into a 5-s epoch. The normal filter option was used as recommended for teenagers (Migueles et al., 2017), and cutoff was used with the Vector Magnitude (Crouter et al., 2015). Time spent in sedentary behaviors and in activity intensities was estimated from raw data. Data captured from 22:30 to 07:00, corresponding to sleep window, were removed from analysis.

Statistical Analysis

An a priori power analysis using the G*Power software package was performed to estimate the needed sample size based on total sleep time data from a similarly designed study (Gu et al., 2020). The effect size in this study was $d = 1.07$, which is a high effect using Cohen's interpretation (Cohen, 1988). With parameters set to be $\alpha = .05$ and power $(1 - \beta) = 0.80$, the required sample size was estimated at $N = 9$ participants, and adding 20% at retention rate, the total sample size is $N = 11$. Data were analyzed with statistical software (R studio). Data are expressed as mean ± *SD*. Descriptive statistics are given as *mean*, *SD*, and range. Paired samples *t* tests were used when data normality was reached (Shapiro–Wilk test) to compare routine dinner (RD) and late dinner (LD) conditions for sleep characteristics, dietary intake at evening meals in Day 3 (D3), mean daily dietary intake of D3, and mean daily dietary intake of the 4 days of both sessions (Session 1, self-selected food vs. Session 2, prepared food). Wilcoxon tests were used if assumption of normality was violated. Cohen's *d* effect sizes (ES) were calculated to determine the magnitude of differences between RD and LD conditions. ES < 0.2, < 0.49 of 0.5 –0.79, and ≥ 0.8 were respectively considered trivial, small, moderate, and large (Cohen, 1988). For Wilcoxon test, rank biserial correlations were converted into Cohen's *d*. Pearson correlation coefficients were used to correlate total sleep time and sleep efficiency with chronotype and subjective sleep quality scores. A value of $p < .05$ was considered statistically significant.

Results

Table 1 summarizes characteristics of the participants. Mean quality of sleep (PSQI) score was 4.2 ± 2.3, with 25% of athletes are poor sleepers. According to the chronotype scale, eight athletes were classified as intermediate type, one as rather evening type, and three as rather morning type. Correlation analyses showed no significant relationship between total sleep time and chronotype and subjective sleep changes ($r = .032$, $p = .920$; $r = .017$, $p = .957$, respectively).

Sleep Data

Sleep outcomes showing significant differences between evening dinnertime conditions are presented in Figure 2. An increase in

Table 1 Participants' Characteristics (*N* = 12 Male)

	Mean ± *SD*	Range
Age (years)	15.8 ± 0.7	14.9–16.6
Anthropometry and body composition		
Height (cm)	177.8 ± 6.1	170–190
Body mass (kg)	81.4 ± 12.3	65.0–102.0
BMI (kg/m^2)	25.8 ± 4.1	22.2–32.6
FM (%)	16.8 ± 6.0	9.8–25.3
FFM (kg)	67.1 ± 5.8	58.1–77.2
Subjective sleep quality (PSQI)	4.2 ± 2.3	0.0–8.0
Chronotype (MESC)	27.2 ± 5.4	16.0–34.0

Note. BMI = body mass index (body mass/height^2); FM = fat mass (skinfold thickness method; Parizkova, 1961); FFM = fat free mass; PSQI = Pittsburgh Sleep Quality Index, scores greater than five at PSQI indicate "poor sleepers;" MESC = Morningness–Eveningness Scale for Children, morning, intermediate, and evening types, equivalent to MESC values of 10–19, 20–32, and 33–42 respectively.

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total sleep time (+24 min, i.e., +6%, $p = .001$, $d = 1.24$) and sleep efficiency (+4.8%, i.e., +6%, $p = .001$, $d = 1.24$) was observed in LD compared to RD. Improvement in sleep efficiency was mainly due to a lower wake after sleep onset (–25 min, i.e., –43%, $p = .014$, $d = -3.20$), a decrease in the number of microarousals (–0.3 events/hr, i.e., –25%, $p = .049$, $d = -0.64$) and awakenings ≥ 90 s (–0.4 events/hr, i.e., –30%, $p < .01$, $d = -0.97$) in LD. There were no significant differences in sleep architecture except for a shorter N3 latency in LD (–6.9 min, i.e., –56%, $p = .03$, $d = -0.778$). Differences in sleep latency failed to reach the significant level ($p = .065$, $d = -0.59$; Table 2).

Dietary Intake Data

The mean daily intake of energy and macronutrient on the measurement days (D0 and D3) was detailed in Table 3. The food

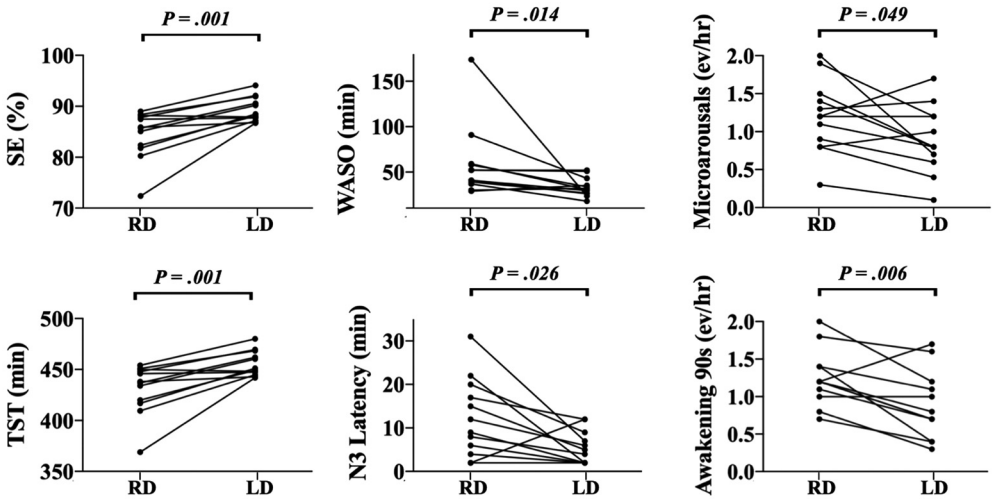

Figure 2 — Sleep efficiency, wake after sleep onset, microarousals, total sleep time, N3 latency, and awakening 90 s in adolescent rugby players ($N = 12$) following RD at 19:00 and LD at 21:00. SE = sleep efficiency; WASO = wake after sleep onset; TST = total sleep time; RD = routine dinner; LD = late dinner; p = significant differences between RD and LD on objective sleep measurements (student paired t test) (Supplementary Material S1 [available online]).

Table 2 Objective Sleep Outcomes Among Junior Rugby Players ($N = 12$) on Baseline (D0) and After 3 Days of Measurement (D3)

	RD (D0)	LD (D0)	Statistics			RD (D3)	LD (D3)	Statistics		
	Mean ± *SD*	Mean ± *SD*	*t*/W	*p*	*d*	Mean ± *SD*	Mean ± *SD*	*t*/W	*p*	*d*
TST (min)	438.6 ± 21.5	439.3 ± 16.4	*26.000*	.919	0.022	431.2 ± 24.3	455.4 ± 12.3	*76.00*	**.001**	1.239
SE (%)	86.0 ± 4.3	86.2 ± 3.2	*26.000*	.919	0.023	84.5 ± 4.8	89.3 ± 2.4	*76.00*	**.001**	1.238
SL (min)	31.8 ± 20.4	26.7 ± 12.9	–0.795	.443	0.286	30.3 ± 12.7	21.0 ± 11.8	–2.049	.065	0.591
WASO (min)	39.6 ± 12.1	42.3 ± 20.7	*30.000*	.839	0.091	58.6 ± 40.0	33.6 ± 10.4	*5.000*	**.014**	3.200
REM (%TST)	20.5 ± 6.7	20.4 ± 6.3	–0.180	.860	0.071	18.3 ± 5.4	20.4 ± 8.8	0.805	.438	0.232
REM (min)	95.2 ± 30.3	89.4 ± 28.1	–1.463	.172	0.423	84.4 ± 26.0	96.0 ± 42.0	0.938	.369	0.271
REM latency (min)	118.0 ± 65.2	141.0 ± 66.6	2.412	.035	–0.857	145.4 ± 80.6	111.1 ± 37.4	–1.585	.141	1.115
N1 (%TST)	4.1 ± 2.2	4.4 ± 2.4	*16.500*	.735	0.179	4.1 ± 3.1	3.8 ± 2.2	*35.00*	.791	0.171
N1 (min)	18.2 ± 10.1	19.0 ± 10.4	33.000	.610	0.172	17.6 ± 12.6	17.2 ± 10.1	38.00	.970	0.056
N2 (%TST)	29.1 ± 10.7	30.4 ± 9.5	–1.018	.331	0.294	29.4 ± 7.6	30.4 ± 11.5	0.557	.589	0.161
N2 (min)	128.5 ± 49.1	133.5 ± 41.8	–0.901	.387	0.260	126.6 ± 33.7	138.5 ± 53.2	1.328	.211	0.384
N3 (%TST)	46.3 ± 10.1	44.9 ± 9.4	–1.031	.325	0.214	48.1 ± 7.5	45.5 ± 11.3	–1.135	.280	0.328
N3 (min)	202.0 ± 41.2	197.5 ± 44.2	–0.499	.627	0.091	207.2 ± 33.0	207.5 ± 53.1	0.007	.995	0.002
N3 latency (min)	9.5 ± 6.7	9.8 ± 5.3	*0.386*	.707	0.036	12.3 ± 9.0	5.4 ± 3.8	7.500	**.026**	0.778
Microarousals (events/hr)	0.9 ± 0.6	1.2 ± 0.9	1.281	.227	0.429	1.20 ± 0.48	0.89 ± 0.44	–2.210	**.049**	0.638
Awakening 30 s (events/hr)	3.5 ± 0.8	3.8 ± 1.0	–0.833	.422	0.241	3.54 ± 1.28	3.84 ± 1.37	*48.50*	.480	0.205
Awakening 90 s (events/hr)	1.0 ± 0.4	1.2 ± 0.4	*1.867*	.089	0.786	1.25 ± 0.37	0.88 ± 0.45	–3.366	**.006**	0.972

Note. Significant differences are in bold values. RD = routine dinner at 19:00; LD = late dinner at 21:00; TST = total sleep time; SE = sleep efficiency; SL = sleep onset latency; WASO = wake after sleep onset; REM = rapid eye movement sleep; N1, N2, and N3 = nonREM sleep stage (N3, deep sleep); t = Student paired t test; W = Wilcoxon test (italicized); p = p value of null hypothesis; d = Cohen's d.

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intakes did not significantly differ in average energy intake on measurement days (D0 and D3) in RD and LD conditions. Macronutrient distribution and intake in g/kg are similar: carbohydrate 45 ± 6% (4.9 ± 1.4 g/kg), protein 17 ± 4% (1.8 ± 0.3 g/kg), and lipid 38 ± 7% (1.9 ± 0.6 g/kg). Likewise, dinner did not differ in energy (RD: 1504 ± 329 kcal vs. LD: 1512 ± 325 kcal, ns).

Physical Activity Data

Young athletes accumulated an average of 2.3 hr of moderate to vigorous physical activity (PA), 5.2 hr of light PA, and 8.0 hr of sedentary behaviors per day. Sedentariness and PA demonstrated no significant differences between RD and LD conditions for the mean of the 5 days nor for the day preceding sleep records (D3; Table 3).

Discussion

In the present study, the effect of time slot between evening dinner and bedtime on sleep quality was investigated in young athletes using a randomized, crossover, within-participant design. As expected, evening dinnertime improved sleep quality of young rugby players. Shifting evening dinner from 3.5 hr before bedtime to 1.5 hr before bedtime increased total sleep time and sleep efficiency, reduced wake after sleep onset, microarousals, awakenings of more than 90 s and N3 latency, and tended to have positive effect on sleep latency during the following night.

The American Academy of Sleep Medicine recommends 8–10 hr of sleep per night for adolescents (Paruthi et al., 2016). In the present study, adolescent rugby players exhibited on average 7 hr of sleep during the night following RD and 7.5 hr of sleep during the night following LD. The population of the present study was suffering from relative sleep deficiency as previously observed (Saidi et al., 2022). A sleep efficiency greater than 85% and a sleep latency shorter than 30 min have also been recommended (Ohayon et al., 2017). In the present study, sleep efficiency moved from 84.5 to 89.3% when evening meal was delayed by 2 hr. Even if differences in sleep latency failed to reach significant level ($p = .065$), a moderate effect size was obtained ($d = -0.59$), and sleep latency decreased by 9 min after the delayed dinner. This lack of significant difference could be due to our sample size.

Only two studies have previously examined the relationship between dinner timing and sleep quality by the questionnaire PSQI (Knufinke et al., 2018) and by accelerometry (Falkenberg et al., 2021) in young adult athletes aged 19–24 years on average. In agreement with these studies, our results confirmed an

Table 3 Dietary Intake, Physical Activity, and Sedentariness in the Two Conditions of Evening Dinnertime on Baseline (D0) and After 3 Days of Measurement (D3)

	RD (D0)	LD (D0)	Statistics		RD (D3)	LD (D3)	Statistics	
	Mean ± *SD*	Mean ± *SD*	*t*/W	*p*	Mean ± *SD*	Mean ± *SD*	*t*/W	*p*
Dietary intake								
Mean daily intake								
Energy (kcal/day)	3,478.2 ± 557.4	3,455.3 ± 579.3	*7.00*	.99	3,447.6 ± 544.8	3,453.1 ± 540.3	*14.00*	.53
CHO (g/kg)	5.0 ± 1.4	4.9 ± 1.4	*5.00*	.99	4.8 ± 1.5	4.8 ± 1.2	*8.00*	.68
Protein (g/kg)	1.9 ± 0.3	1.9 ± 0.3	*1.00*	.20	1.6 ± 0.4	1.6 ± 0.4	*11.00*	.99
Lipid (g/kg)	1.8 ± 0.7	1.8 ± 0.7	*5.00*	.99	1.9 ± 0.5	1.9 ± 0.5	*7.00*	.53
CHO (%)	45.9 ± 6.3	45.3 ± 6.7	*7.00*	.58	44.6 ± 6.3	44.6 ± 6.5	*8.00*	.99
Sugar (%)	13.3 ± 3.9	13.2 ± 3.6	*5.00*	.99	15.4 ± 5.2	15.4 ± 5.1	*10.00*	.99
Protein (%)	17.6 ± 2.7	17.8 ± 2.5	*4.00*	.86	15.6 ± 4.6	15.6 ± 4.5	*10.00*	.59
Lipid (%)	36.5 ± 8.0	36.9 ± 8.1	*3.00*	.58	39.8 ± 5.9	39.8 ± 6.0	*7.00*	.99
Water (L/day)	2.0 ± 1.0	2.0 ± 1.0	*2.00*	.79	3.8 ± 1.1	3.8 ± 1.1	*1.00*	.99
Evening dinner								
Energy (kcal/day)	1,472 ± 251.5	1,485.8 ± 296.3	*3.00*	.37	1,536.6 ± 401.4	1,537.8 ± 383.2	*3.00*	.99
CHO (g/kg)	1.7 ± 0.4	1.8 ± 0.4	*1.00*	.99	1.9 ± 0.76	1.9 ± 0.70	*3.00*	.99
Protein (g/kg)	0.9 ± 0.3	0.9 ± 0.3	*0.00*	.99	0.8 ± 0.3	0.8 ± 0.3	*18.00*	.20
Lipid (g/kg)	0.9 ± 0.3	0.9 ± 0.3	*0.00*	.37	1.0 ± 0.4	1.0 ± 0.4	*1.00*	.99
CHO (%)	38.1 ± 4.3	37.9 ± 4.0	*2.00*	.99	39.0 ± 9.1	39.1 ± 9.0	*2.00*	.79
Sugar (%)	10.7 ± 3.4	10.9 ± 3.5	*1.00*	.99	11.6 ± 5.2	11.7 ± 4.9	*2.00*	.79
Protein (%)	19.4 ± 5.8	19.4 ± 5.8	*3.00*	.37	18.4 ± 9.6	18.3 ± 9.6	*5.00*	.41
Lipid (%)	42.5 ± 8.9	42.8 ± 8.6	*1.00*	.99	42.6 ± 9.8	42.6 ± 9.7	*3.00*	.99
Water (L/day)	0.9 ± 0.4	0.9 ± 0.4	*1.00*	.99	1.3 ± 0.5	1.2 ± 0.5	*1.00*	.99
Physical activity								
SB (hr)	7.3 ± 0.9	7.2 ± 0.7	0.50	.63	8.2 ± 2.3	8.1 ± 1.5	0.16	.88
LPA (hr)	5.6 ± 0.5	5.7 ± 0.5	–0.74	.48	5.1 ± 1.4	5.3 ± 1.1	–0.57	.58
MVPA (hr)	2.6 ± 0.7	2.6 ± 0.7	*44.50*	.69	2.2 ± 1.0	2.1 ± 0.8	0.28	.79

Note. Data processing carried out during the active phase, from 07:00 to 22:00 (over 15.5 hr). CHO = carbohydrate; RD = routine dinner at 19:00; LD = late dinner at 21:00; D0 = Day 0; D3 = Day 3; SB = sedentary behavior; D; *t* = student paired *t* test; W = Wilcoxon test (italicized); *p* = *p* value of null hypothesis; *d* = Cohen's *d*; LPA = light physical activity; MVPA = moderate to vigorous physical activity.

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improvement of the total sleep time (+24 min, +6%) and sleep efficiency (+6%) in the delayed dinner condition. Sleep efficiency improvement was mainly due to a lower wake after sleep onset (−43%), a decrease in the number of microarousals (−25%), and of awakenings ≥90 s (−30%) in the delayed condition. Only the improvement of wake after sleep onset contrast from Falkenberg's study, which observed a small decrease with the delay reduction between the consumption of the last meal and bedtime (~2 min for each additional hour). The paucity of data in the literature makes it is difficult to precisely understand the relationships between dinner timing and sleep in athletes. The current findings on N3 latency are in accordance with the literature. Indeed, in the present study, N3 latency was shortened by approximately 56% in late dinner condition. Duan et al. (2021) performed electroencephalogram (EEG) spectral analyses in 26-year-old healthy volunteers and demonstrated that moving dinner from 5 to 1 hr closer to the sleep period induced deeper sleep at the beginning of the night.

Evening dinner must provide sufficient energy for the night to keep the body and brain functioning during this fasting period. For example, during rapid eye movement sleep, the brain consumes as much glucose and oxygen as during wakefulness (Benoit & Foret, 1995). If the delay between dinner and bedtime is too long, orexigenic signals would be activated in the brain, promoting wakefulness and consequently inducing fragmented sleep and awakenings (Falkenberg et al., 2021). Indeed, different nervous and hormonal signals are involved in fasting/feeding to ensure energy homeostasis (Ruddick-Collins et al., 2020) throughout the 24 hr of the day. Many of them play a crucial role in the regulation of sleep and wakefulness. For example, ghrelin secreted by stomach cells stimulates feeding via activation of orexin-producing neurons which are implicated in maintaining wakefulness (Toshinai et al., 2003). In contrast, leptin produced by adipose cells inhibits orexigenic neurons producing neuropeptide Y and agouti-related protein in order to counteract ghrelin effects and increases satiety (Utriainen et al., 1996). Therefore, sleep can be promoted either by inhibiting the mechanisms that promote wakefulness or by increasing the factors that promote sleep. Indeed, in athletes, the uptake of glucose by the muscles is increased due to the greater presence of glucose transporter type 4 (GLUT4) in the skeletal muscles. As a result, postprandial blood glucose levels decrease. Athletes would therefore have higher fasting ghrelin levels than inactive individuals (Dorling et al., 2018). Thus, it can be hypothesized that in young athletes, if the time between evening dinner and bedtime is too long, orexigenic signals could be activated and lead to shorter and poorer quality sleep.

Another hypothesis that could explain the improvement in sleep quality would be related to the phenomenon of postprandial sleepiness. Neurotransmitters such as serotonin are associated with the sleep–wake cycle and postprandial sleepiness. In fact, the ingestion of carbohydrate promotes the secretion of insulin which decreases plasma levels of neutral amino acids and promotes the passage of tryptophan across the blood–brain barrier. As a result, serotonin synthesis is increased, favoring the onset of sleep and the release of melatonin. Even if mechanisms of postprandial sleepiness might be similar in both conditions, delaying evening dinnertime presumably induced a delay in the kinetics of such mechanisms. In addition, Duan et al. (2021) suggested that postprandial sleepiness would help to achieve the deep sleep that is essential for body recovery and growth. It could therefore be hypothesized that reaching deep sleep more quickly would be beneficial to athletes' recovery.

This study is the first to examine the effect of delayed evening meal on sleep quality in young athletes. One of the strengths is the randomized, crossover design carried out under usual living conditions in boarding school. Another strength is the objective and validated methods used for sleep, physical activity, and food intake measurements. To limit our intervention to evening dinner timing, participants were requested to maintain identical dietary intakes, PA, training, and sleep–wake schedules throughout the study. Thus, improvement in quality of sleep can be attributed to neither changes in quality or quantity of food ingested between conditions as previously mentioned by St-Onge et al. (2016) nor changes in PA schedule between conditions as previously showed in different protocols conducted in good or poor sleepers (Chennaoui et al., 2015). However, one limit of the present study is that the parameters explaining mechanisms involved, such as hunger and satiety and digestive discomfort, were not assessed. Finally, although the number of participants remains small, potentially limiting the interpretation of findings to other sports or populations, the sample is homogeneous, and the retention rate of participants is good.

Conclusion

Shifting from a routine dinner at 19:00 to a late dinner at 21:00 had positive effects on objective measures of sleep in young rugby players. An evening mealtime 1.5 hr before bedtime leads to less fragmented sleep and therefore improves sleep quality in young athletes. The fact that a simple modification in the timing of evening meals can significantly improve sleep quality lends itself to an easy, feasible, and convenient strategy for promoting sleep in young athletes. Given the crucial role of sleep for young athletes' health and performance, future research is needed to develop nutritional strategies that would be easy to implement to improve sleep.

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